

Zemax

[illegible]

Date	Version	OpticStudio version	Comment
2019/12/10	1.0	19.8	Creation
2020/09/20	1.1	20.2	Updates for OpticsTalk + Envision. Cleaning up the output. Additionally,

Code Exchange

Zemax

			a reformatted version has been made available which splits the calculation into child/parent macros.
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Instructions

1. Installation

Copy the macro to the {Zemax}\Macros folder on your machine.

2. How to use

In OpticStudio, navigate to **Programming...Edit/Run** and open the macro for editing. Modify lines 6, 7, and 8. These correspond respectively to the number of steps across the pupil, the surface of interest, and the wavelength of interest.

```

1 !!!!!!!!!!!!!!!!!!!!! Step 1 !!!!!!!!!!!!!!!!!!!!!
2 ! Choose the surface
3 ! Choose number of rays to trace in X and Y across the pupil
4 ! Choose wavelength of interest
5
6 numSteps = 10
7 surf = 5
8 wave = 2
9

```

On line 56 we define which optimization operand we will use for this analysis. By default, we are using RAID. If the operand is updated, you must make sure line 60 is still valid. Line 60 represents the operand inputs. For the RAID operand, we input the following:

```
rayData = OPEV(angle, surf, wave, hx, hy, py, px)
```

Once the values of the macro are confirmed, save it and run. The output will be a field-by-field analysis of the optimization operand at different pupil locations on a given surface. The text can be directly output to an external file, or copied from the Text Viewer into a post-processing program.

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